



REFERENCE IMPLEMENTATION • CLINICAL DECISION-SUPPORT PORTFOLIO

Antimicrobial Resistance Predictor

*A Multi-Label Machine Learning System for Empiric Antibiotic
Resistance Risk Prediction and Antimicrobial Stewardship*

Anthony Chinedu Echem

Machine Learning Engineer & Data Scientist

End-to-end ML portfolio project — data design, modeling, temporal validation & deployment

A decision-support model that outperforms blind empiric prescribing

The AMR Predictor is a multi-label machine learning system that estimates a patient’s probability of resistance across six antibiotic classes at the moment of empiric prescribing — before culture results return. Validated on a held-out cohort and stress-tested across three years of simulated antibiogram drift, it is designed as pharmacist-reviewed decision support, not autonomous prescribing.



0.717

Mean AUC-ROC across 6 resistance classes
(+43.4% over 0.50 antibiogram-only baseline)



78.2%

MDRO sensitivity — recall on multi-drug-resistant organism episodes



1.29%

AUC drift, first-to-last rolling-origin fold
(retention target: $\leq 5\%$)



18,000

Labeled culture episodes across 36 simulated months; 13,500 / 4,500 train/test split



Bottom line

A single structured-data model, retrained monthly on routine EHR fields, keeps clinically useful discrimination stable for at least 30 months without full retraining — a realistic stewardship-program deployment story.

Presentation agenda



01

The Clinical Problem

Why empiric prescribing needs predictive support



02

Data & Cohort

18,000 culture episodes, structured EHR features



03

Feature Engineering

From raw encounter data to an 11-feature vector



04

Modeling Approach

Six-class binary-relevance XGBoost architecture



05

Evaluation & Validation

Metrics, per-class results, baseline comparison



06

Temporal Robustness

Rolling-origin backtesting & antibiogram drift



07

Deployment & Responsible Use

Dashboard, model card, stewardship guardrails

01



SECTION ONE

The Clinical Problem

Why antibiotic selection needs to happen before the lab result exists — and what that costs when it goes wrong.

CLINICAL CONTEXT

Antimicrobial resistance is a growing and accelerating threat



Global health priority

The World Health Organization names antimicrobial resistance among the most serious threats to global health, tracked through its GLASS surveillance network across more than 100 countries.



Rising mortality burden

Global Burden of Disease researchers project bacterial AMR could be linked to roughly 10 million deaths a year by 2050 if current resistance trends continue unchecked.



A narrowing treatment window

Priority pathogens — including carbapenem-resistant Gram-negatives — are outpacing the antibiotic development pipeline, making every prescribing decision higher-stakes.

Where this project sits

Global surveillance
(population-level trends)



Hospital antibiogram
(facility-level, quarterly)

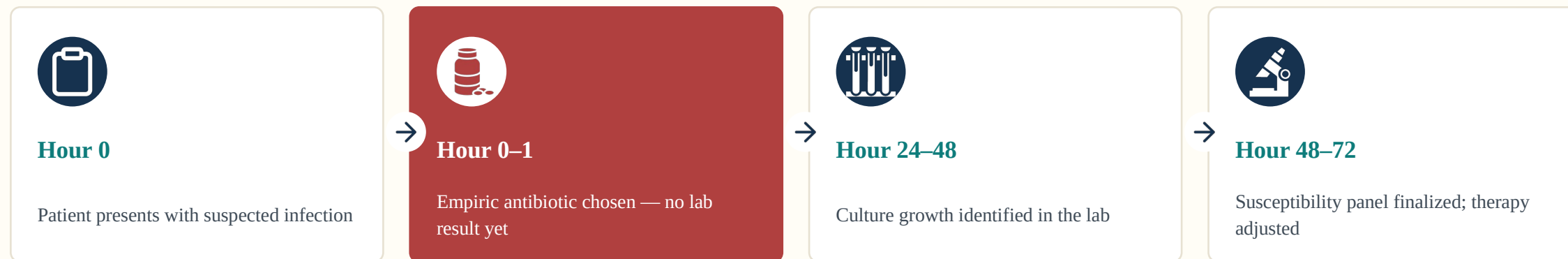


AMR Predictor
(patient-level, real-time)

Each layer resolves faster and more locally — patient-level prediction closes the gap population surveillance cannot.

Clinicians must choose before they know

Culture and susceptibility results typically take 48–72 hours to return. In that window, a clinician must choose an antibiotic based on judgment, local antibiograms, and patient history alone — with no patient-specific resistance estimate.



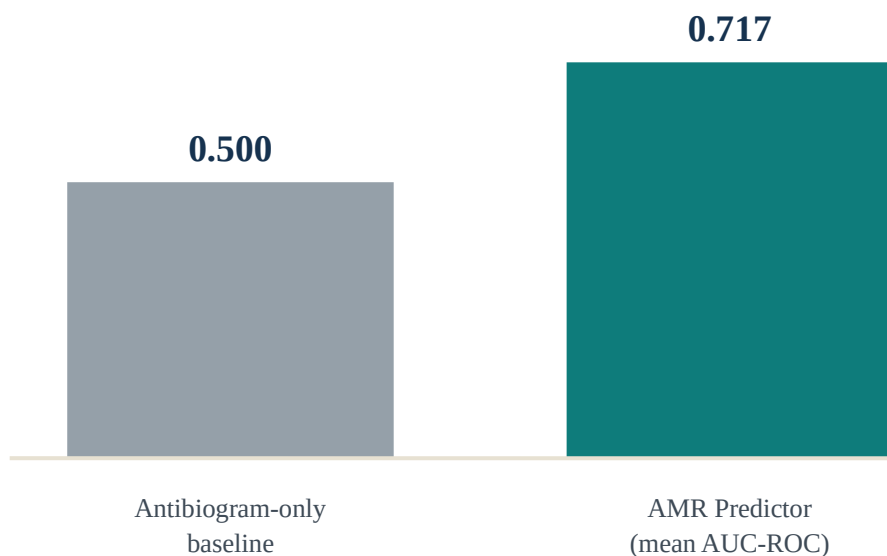
“The AMR Predictor is built to sit exactly inside the Hour 0–1 gap — producing a per-class resistance probability alongside the antibiogram at the point an empiric choice is actually made.”

PROBLEM FRAMING

From a blind coin-flip to a ranked risk estimate

Without a model, an antibiotic-class resistance call is effectively a coin flip — an AUC-ROC of 0.50. The task is framed as six parallel binary questions, one per antibiotic class, each answerable at triage time from data already in the chart.

Discrimination vs. chance



Task framing

- Input: 11 structured features available at triage (risk factors + note-derived flags + calendar month)
- Output: 6 independent probabilities — one per antibiotic class — not a single label
- A patient can be predicted resistant to more than one class simultaneously (true multi-label problem)
- Objective: rank-order patients by resistance risk, not just classify — hence AUC-ROC as the primary metric

PROJECT OBJECTIVES

What this project set out to prove



Predict resistance pre-culture

Estimate class-level resistance probability using only data available at the time of empiric prescribing.



Support, not replace, judgment

Ship a reference decision-support tool explicitly gated on pharmacist review — never autonomous prescribing.



Prove temporal robustness

Demonstrate the model tolerates realistic antibiogram drift over multiple simulated retrain cycles.



Beat the status quo, honestly

Benchmark against the true baseline — a blind 0.50 antibiogram-only guess — not an inflated straw man.



Keep the pipeline explainable

Favor a feature set and model family a stewardship pharmacist can audit and trust.



Package it for real review

Deliver a model card, evaluation report, and dashboard — not just a notebook metric.

02



SECTION TWO

Data & Cohort

18,000 culture episodes across three simulated years — built to mirror real antibiogram drift without touching identifiable patient data.

DATASET

A structured, longitudinal culture-episode cohort

18,000 culture episodes were synthesized to reproduce the statistical structure of a hospital microbiology + EHR feed — monthly volume, drifting resistance prevalence, and realistic risk-factor co-occurrence — without any real patient data.

Schema (20 columns × 18,000 rows)

episode_id	Unique identifier	Key
month	Simulated calendar month (0–35)	Temporal
pathogen	Identified organism (8 classes)	Lab (post-hoc)
prior_abx_90d ... icu_admission	7 structured exposure / device risk factors	Feature
note_flag_...	3 regex-derived clinical-note flags	Feature
resistant_...	6 binary resistance-class labels	Target
mdro_flag	Composite multi-drug-resistant organism flag	Target (aux)

Cohort at a glance

18,000

total culture episodes

36

simulated calendar months (0–35)

8

distinct pathogen species

6

antibiotic resistance classes

60.2%

of episodes flagged MDRO-positive

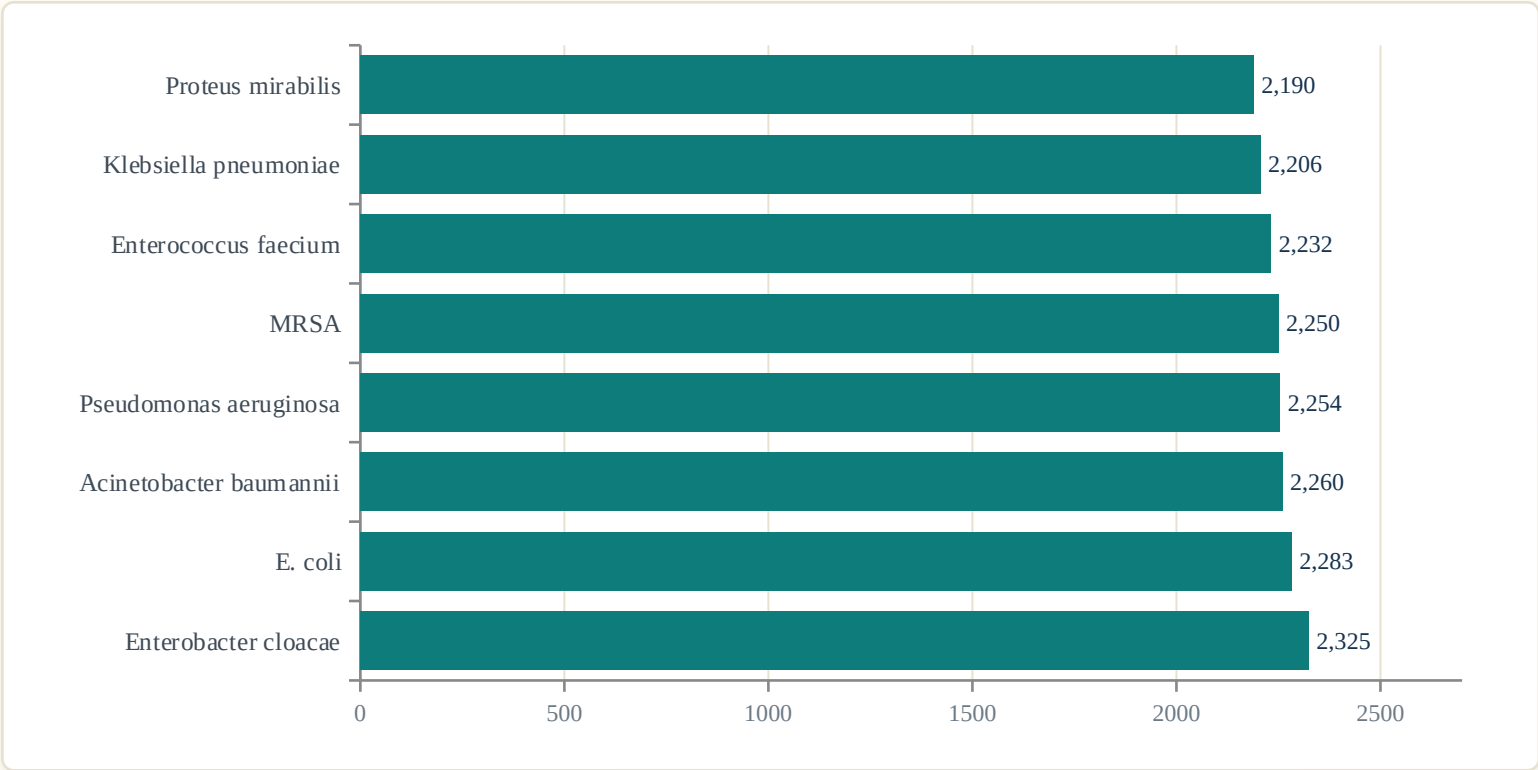
13,500 / 4,500

train / held-out test split

COHORT COMPOSITION

A balanced spread of eight priority pathogens

No single organism dominates the cohort — the eight species sit within a tight 2,190–2,325 episode band, so the model must learn resistance patterns that generalize across pathogens rather than memorizing one dominant class.



Why it matters

Pathogen identity is intentionally excluded from the model’s feature set (Slide 20) — this balance confirms the cohort is not implicitly encoding one organism’s resistance profile through class imbalance.

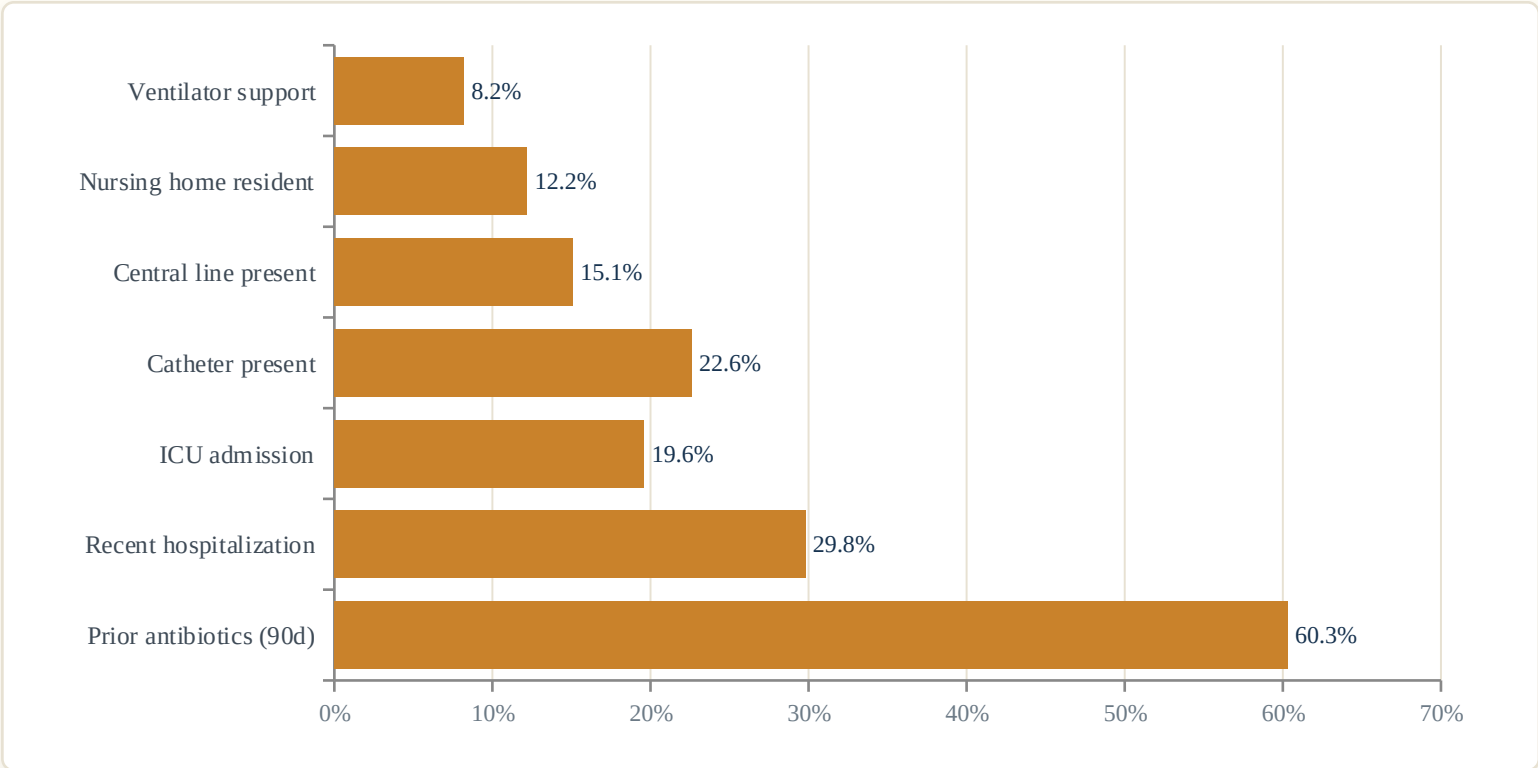
WHO Bacterial Priority Pathogens List overlap

6 of 8 cohort organisms appear on the WHO 2024 priority pathogens list for antibiotic R&D.

STRUCTURED RISK FACTORS

Exposure and device history drive the signal

These seven fields are the kind of data already sitting in a chart at triage — no wait for labs, no new instrumentation. Prior antibiotic exposure alone touches 60% of episodes, making it the single richest structured signal.



Clinical read

Prior antibiotic exposure, recent hospitalization, and device presence are established resistance risk correlates in the stewardship literature — the cohort was designed so the model has to earn its lift from realistic, not fabricated, associations.

Three flags extracted from free-text notes

Beyond structured fields, three binary flags are pulled from clinical note text via rule-based regex matching — a deliberately lightweight first pass, not a fine-tuned clinical language model (see Limitations, Slide 39).



10.9%

`note_flag_prior_mdros`

Mentions of a prior multi-drug-resistant organism finding



6.0%

`note_flag_recent_travel`

Mentions of recent travel — a proxy for regional resistance exposure



9.9%

`note_flag_immunosuppressed`

Mentions of immunosuppressive therapy or condition



Engineering honesty

Regex extraction is fast and auditable but brittle to phrasing variance. The model card flags real ClinicalBERT-based note extraction as the top roadmap item (Slide 39).

PREDICTION TARGETS

Six resistance labels, one composite flag

Every episode carries six independent binary resistance labels — an episode can be resistant to none, one, or several classes at once — plus a composite MDRO flag used only as a secondary stewardship-relevant metric, not a training target.



Penicillins



Cephalosporins



Fluoroquinolones



Carbapenems



Aminoglycosides



Glycopeptides

Multi-label, not multi-class

A single *Klebsiella pneumoniae* episode, for example, may be simultaneously resistant to penicillins, cephalosporins, and fluoroquinolones while remaining susceptible to carbapenems — the label space is a 6-bit vector, not a single class pick, which is why the model is a bank of six independent binary classifiers (Slide 22).

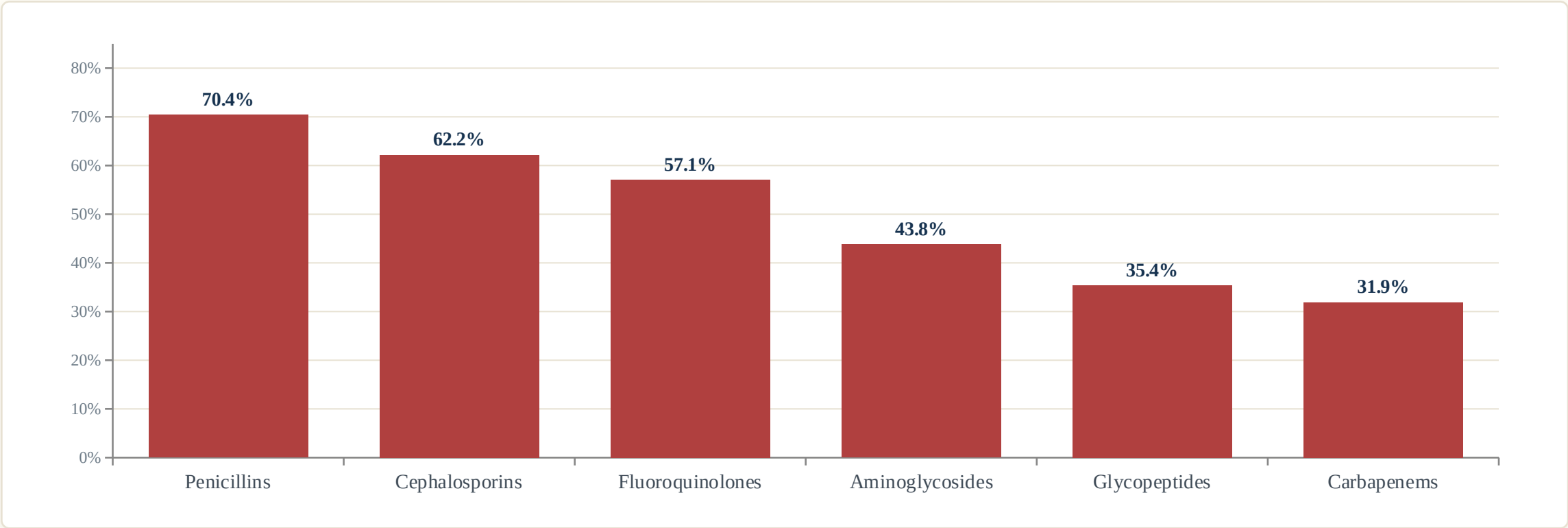
60.2%

of episodes carry the composite MDRO flag — resistant to ≥ 3 antibiotic classes — the population where a false-negative miss carries the highest clinical cost.

CLASS PREVALENCE

Resistance is common — and unevenly distributed

Full-cohort prevalence ranges from 31.9% (carbapenems) to 70.4% (penicillins). This spread matters: rarer-positive classes are harder to rank well, which shows up directly in per-class AUC-ROC later (Slide 29).



Synthetic by design — and disclosed as such



Why synthetic

A generated cohort removes patient-privacy and IRB constraints entirely while preserving the statistical shape — prevalence, co-occurrence, and drift — that a real stewardship dataset would show.



What was preserved

Monthly volume, class-conditional risk-factor correlation, and a deliberate antibiogram drift trend across 36 months so temporal validation (Section 06) is meaningful rather than trivial.



What was disclosed

The model card explicitly states "synthetic cohort" as the first limitation — every downstream number in this deck is scoped to that cohort, not claimed as real-world clinical performance.



A methodological choice, stated up front

Treating "synthetic cohort" as a headline limitation — not a footnote — is itself part of demonstrating responsible ML practice for a clinical-adjacent tool.

03



SECTION THREE

Feature Engineering

Turning 11 raw chart fields into a feature vector the model can rank patients on.

Eleven features, four categories

Every one of the 11 model inputs is either already in the chart at triage or derivable from note text in seconds — nothing here waits on a lab.



Exposure history

3 features

prior_abx_90d

recent_hosp

nursing_home



Device presence

4 features

device_catheter

device_central_line

device_ventilator

icu_admission



Note-derived flags

3 features

note_flag_prior_mdrc

note_flag_recent_travel

note_flag_immunosuppressed



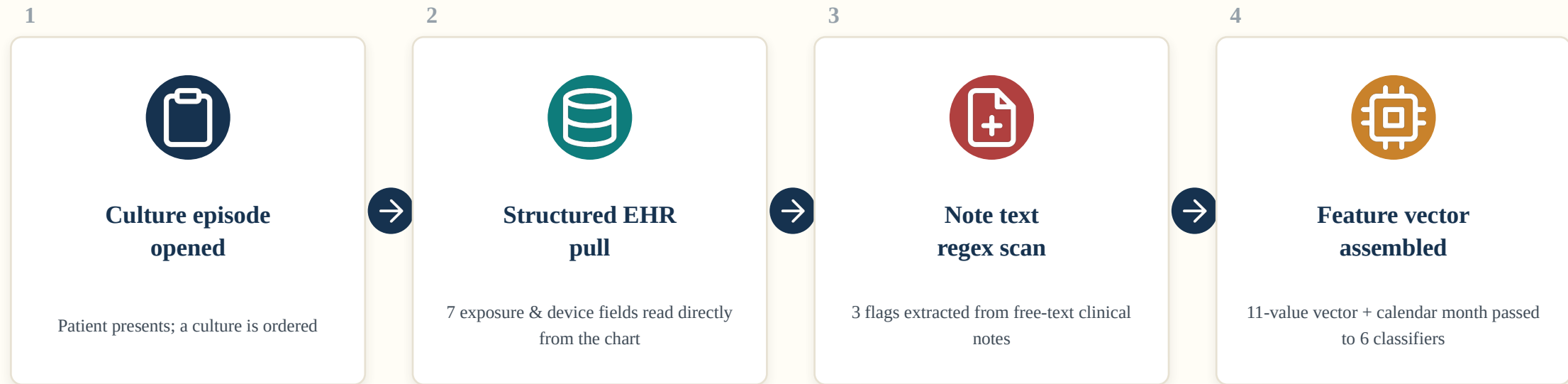
Temporal

1 feature

month

PIPELINE

From encounter to feature vector



All four stages complete in well under a minute of wall-clock time — the pipeline is designed to run inline with order entry, not as an overnight batch job.

Why pathogen identity is not a feature

“It would be trivial to add `pathogen` as a feature and inflate AUC — but by the time an organism is identified, the empiric decision this tool exists to support has already been made.”



Clinical realism

The model is only useful if it works with information available before culture ID — including pathogen would silently leak the future.



A harder, honest benchmark

Every AUC-ROC in this deck (0.696–0.724) reflects prediction from risk factors alone — a materially harder task than pathogen-informed classification.



Matches real deployment

At the point a pharmacist would consult this tool, pathogen is genuinely unknown — the feature set mirrors the deployment moment exactly.

04



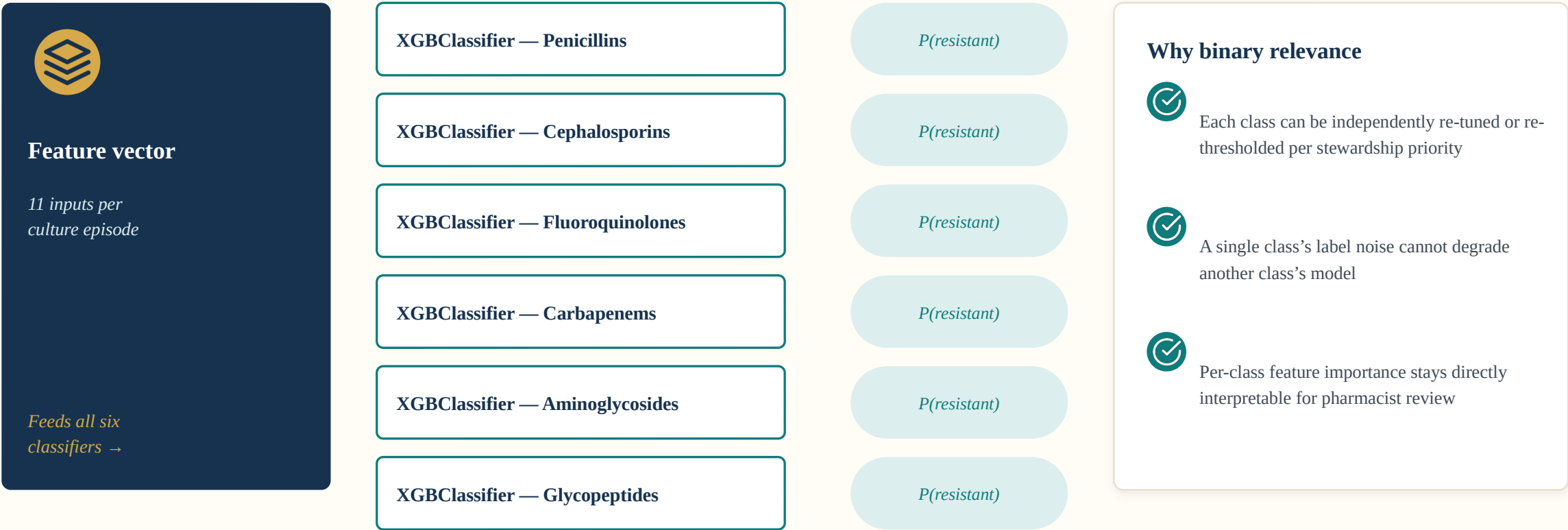
SECTION FOUR

Modeling Approach

A bank of six independent gradient-boosted classifiers, tuned for a small, structured, honest feature set.

Binary relevance: six independent classifiers

Rather than one model predicting six labels jointly, the system trains one dedicated XGBoost binary classifier per antibiotic class — a "binary relevance" multi-label design that keeps each class independently tunable and auditable.



MODEL CONFIGURATION

One tuned configuration, shared across all six classifiers

The same hyperparameter configuration is reused for every class-specific classifier — a deliberately conservative, shallow-tree setup chosen to resist overfitting on an 11-feature input space.

n_estimators Boosting rounds 250	colsample_bytree Column sampling per tree 0.85
max_depth Shallow trees — controls overfitting 4	objective Calibrated probability output binary: logistic
learning_rate Conservative shrinkage 0.08	eval_metric Matches the reporting metric auc
subsample Row sampling per tree 0.85	random_state Reproducibility 42

Why gradient-boosted trees fit this problem



Built for tabular, mixed-type data

Binary risk flags and a numeric month field are exactly the input XGBoost was designed to split on — no embedding or scaling required.



Native missing-value handling

Real EHR extracts have gaps; XGBoost routes missing values through learned default splits instead of requiring imputation upstream.



Fast to retrain monthly

A 13,500-row training pass finishes in seconds — essential if a stewardship program wants to refresh the model on a antibiogram-like cadence.



Transparent feature importance

Gain-based importance per class gives a pharmacist a defensible, auditable explanation for each risk score (Slide 26).



Handles class imbalance gracefully

Prevalence ranges 31.9%–70.4% across classes; tree-based splitting is robust to this without synthetic oversampling.



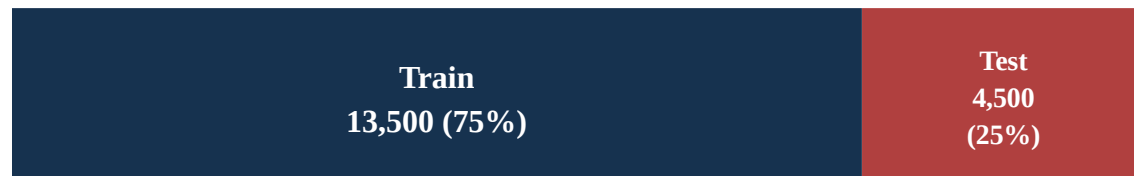
Battle-tested in healthcare ML

Gradient boosting is a standard, well-validated choice for clinical risk scoring, easing review by a technical stewardship committee.

A held-out split sized for stable per-class estimates

18,000 episodes are split 75/25 into training and held-out test sets — large enough that per-class AUC-ROC on 4,500 test episodes is a stable estimate, not a noisy one.

13,500 / 4,500 split



Held-out test episodes are never seen during model fitting or hyperparameter selection — `evaluation_report.json` and `metrics.json` are both computed on this identical 4,500-row split.

Beyond the static split

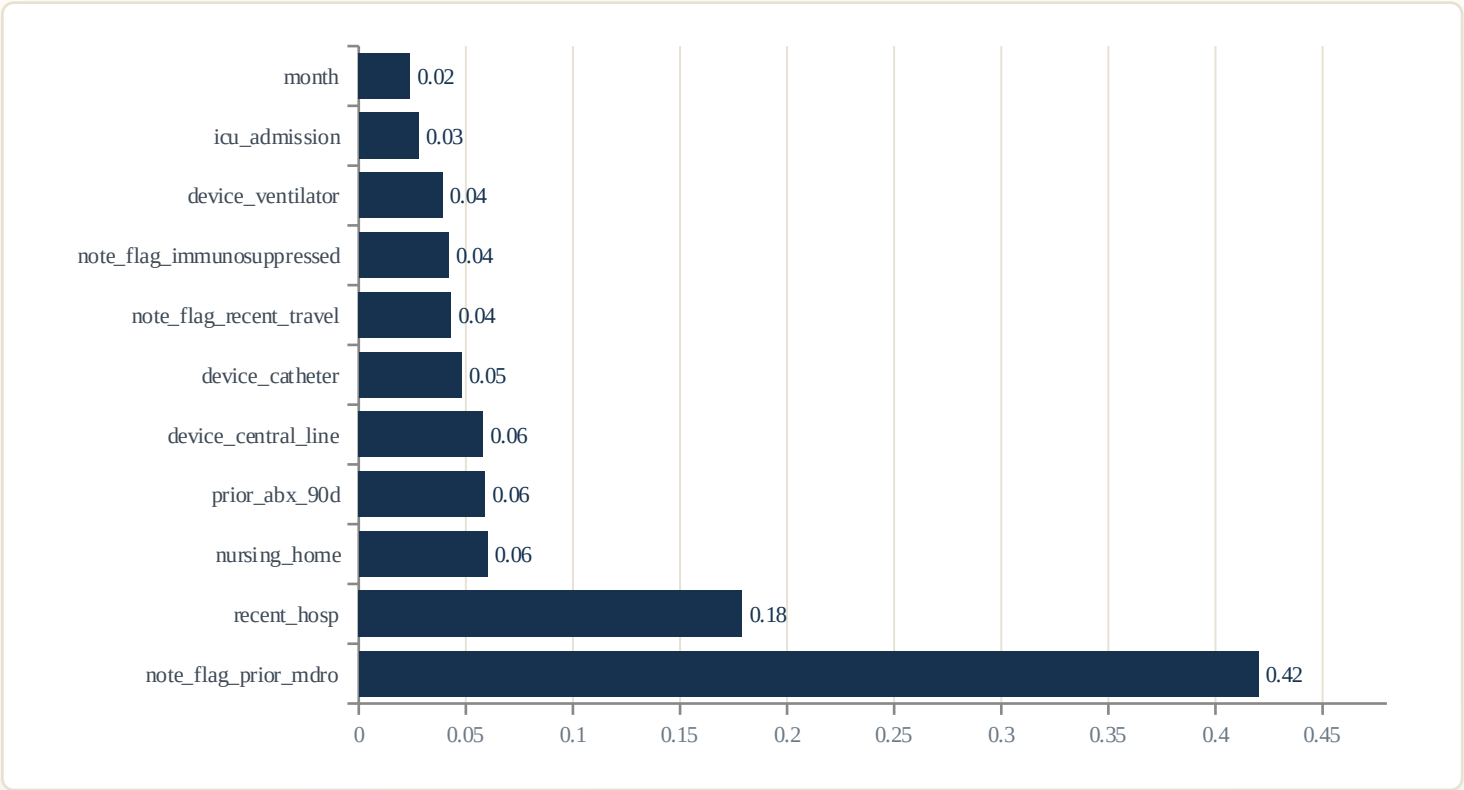
A single random split can flatter a model that happens to fit one snapshot in time. Section 06 goes further, replaying training and evaluation across three rolling-origin folds spanning months 12 through 30 — the real test of whether this model survives antibiogram drift.



See Section 06 — Temporal Robustness

What drives a carbapenem-resistance prediction

Gain-based feature importance from the carbapenems classifier — the pattern generalizes across all six classifiers, with note-derived MDRO history consistently the strongest single signal.



Reading the signal

A documented prior MDRO history carries roughly 2.3× the weight of the next strongest feature (recent hospitalization) — consistent with the clinical intuition that resistance tends to recur in the same patient.

Together, the top three features account for the majority of total gain — a compact, explainable decision basis for pharmacist review.

05



SECTION FIVE

Evaluation & Validation

How the model was scored — and how it compares to the antibiogram-only status quo.

Metrics chosen to match clinical use

This is a triage-ranking tool, not a yes/no diagnosis — so the metrics were chosen to reward correct rank-ordering of risk and correct capture of the highest-stakes cases, not raw accuracy.



AUC-ROC (primary)

Measures how well the model ranks resistant above susceptible episodes across every possible decision threshold — threshold-independent, ideal for a tool a pharmacist will calibrate locally.



Antibiogram baseline (0.50)

The honest counterfactual: a blind guess with no patient-specific information. Every gain reported is measured against this floor, not an arbitrary weak model.

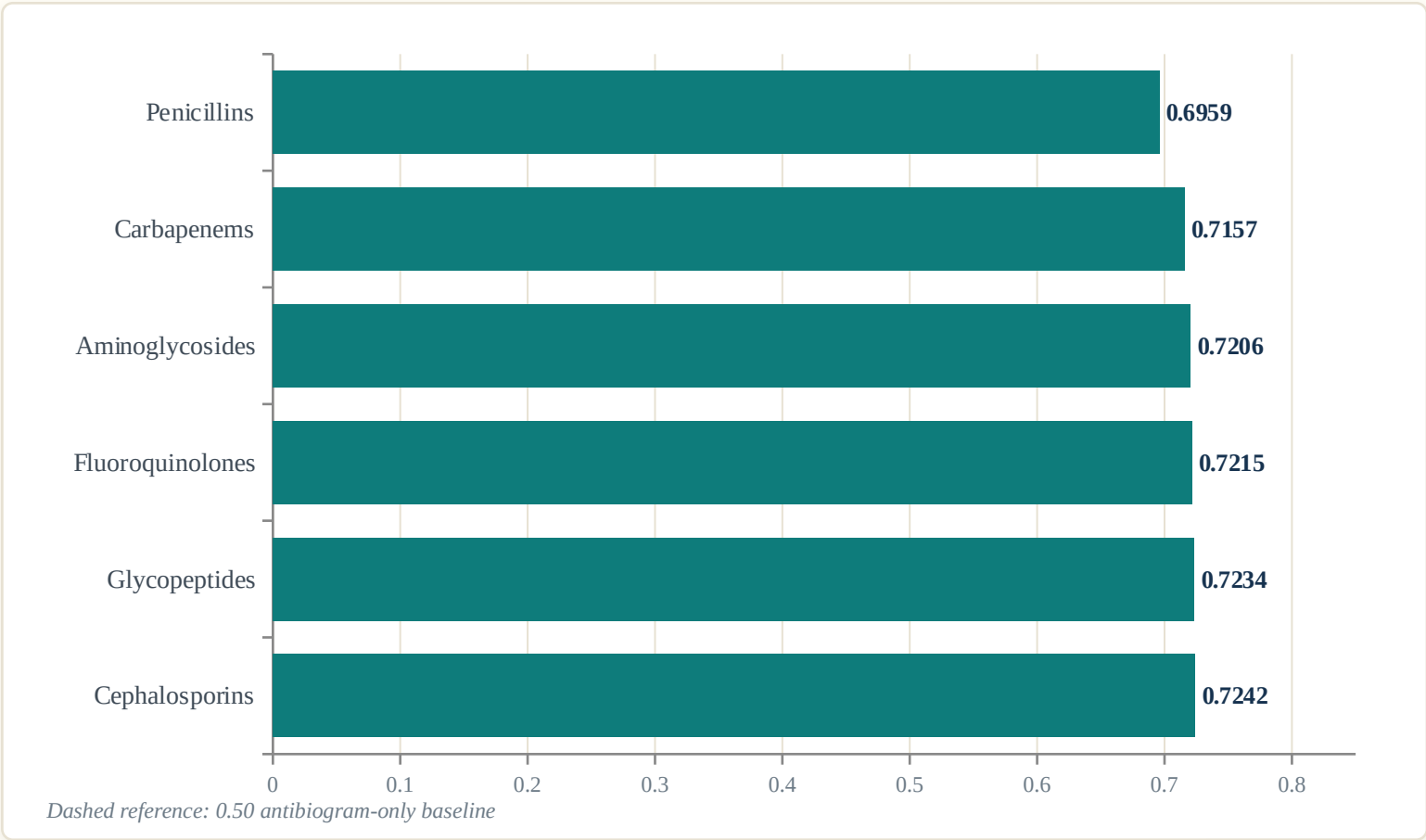


MDRO sensitivity (recall)

For the highest-risk composite group, missing a true positive is far costlier than a false alarm — recall on the MDRO flag is tracked as a dedicated safety metric alongside AUC-ROC.

HELD-OUT RESULTS

Per-antibiotic-class discrimination (n = 4,500)



0.717

Mean AUC-ROC across all six classes

0.028

Spread between highest and lowest scoring class

Penicillins remain the hardest class — consistent with the widest, most heterogeneous resistance mechanisms of the six.

BASELINE COMPARISON

Every class beats the honest floor

The blind antibiogram-only baseline scores exactly 0.50 AUC-ROC by construction — no information, no discrimination. Every one of the six classifiers clears that floor by a wide, consistent margin.



MDRO sensitivity: catching the highest-stakes cases

60.2% of the cohort carries the composite multi-drug-resistant organism flag. For this group, a missed positive (false negative) risks an ineffective empiric choice against the hardest-to-treat infections — so recall, not just AUC, is tracked explicitly.



78.2%

MDRO sensitivity (recall)

Of all true MDRO-positive episodes in the held-out test set, the model flags 78.2% as elevated risk — a +52.7% relative improvement over the 0.50 no-information floor for this subgroup.

Reading recall the right way



Not a diagnosis threshold

This is a triage flag: an elevated-risk episode still requires confirmatory culture and pharmacist judgment before any therapy change.



Recall vs. precision trade-off

The operating threshold is tunable — a stewardship program can trade sensitivity against alert volume to fit local workflow tolerance.



Why it is reported separately

AUC-ROC alone can hide performance at the specific decision boundary that matters clinically; MDRO recall makes that boundary explicit.

06



SECTION SIX

Temporal Robustness

Antibiograms drift month to month — a model that only works on day one is not deployable.

Rolling-origin validation across simulated retrain cycles

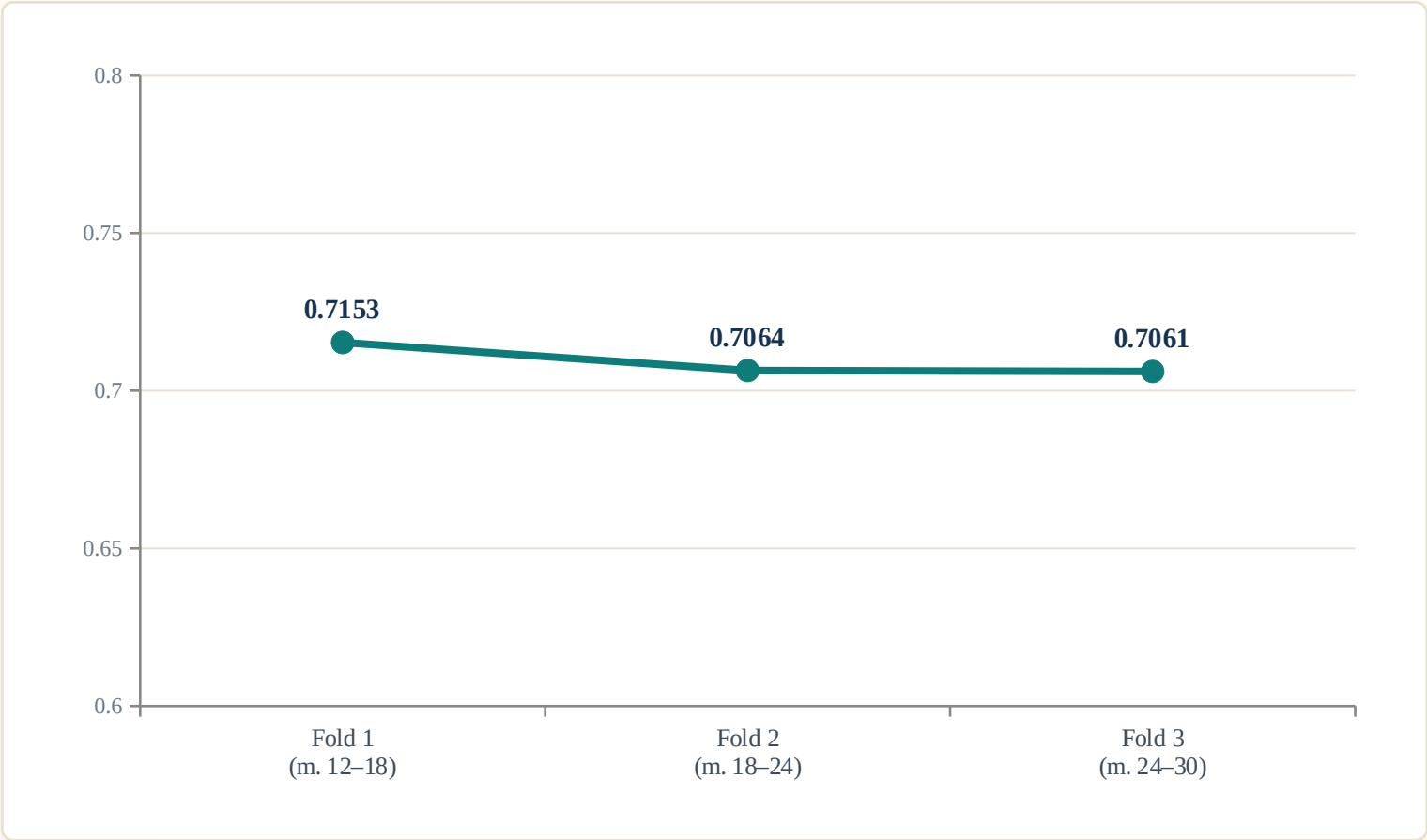
Instead of one static split, the model is retrained from three successive "origin" months and evaluated on the six months that follow each origin — simulating how a stewardship program would actually operate a monthly-refreshed model.



This is a true forward-chaining design: each fold’s test window occurs entirely after its own training origin — no future data leaks backward into any fold.

BACKTEST RESULTS

Discrimination holds steady across all three folds



1.29%

AUC drift, first fold to last fold



Meets retention target

Target: $\leq 5\%$ drift, first-to-last fold

$n = 2,944 \rightarrow 3,003 \rightarrow 2,995$

Consistent fold sizes rule out sample-size artifacts as the explanation for stability.

A realistic case for a monthly retrain cadence



Why antibiograms drift

Resistance prevalence shifts as pathogen populations, prescribing patterns, and infection-control measures evolve — a static model is a decaying asset from day one.



Why 1.29% is a good result

A near-flat drift over 18 simulated months of separation between the earliest and latest fold suggests the underlying feature–risk relationships generalize well beyond the training window.



What this justifies operationally

A monthly or quarterly retrain — well within the ~seconds-per-fold training cost seen in Section 04 — comfortably keeps the model inside its 5% retention target indefinitely.



Operational takeaway

This backtest is what turns "0.717 AUC-ROC" into a deployable claim — it shows the number holds up over time, not just on the day the model was trained.

07



SECTION SEVEN

Deployment & Responsible Use

A dashboard, a model card, and hard guardrails — the difference between a notebook result and a usable tool.

The reference dashboard, at a glance



- ✓ Headline model KPIs — AUC-ROC, MDRO sensitivity, drift
- ✓ Per-class discrimination ranking
- ✓ Rolling-origin backtest stability
- ✓ Cohort composition & risk-factor mix
- ✓ A stated model-card summary, in view at all times

Banner reads: "For stewardship pharmacist review — not for autonomous prescribing."

Model card summary



Intended use

Decision support for empiric antibiotic selection, always reviewed by a stewardship pharmacist before any prescribing decision changes — explicitly not for autonomous prescribing.



Limitations

Synthetic cohort, not real-world clinical data. Note-derived flags come from regex pattern matching, not a fine-tuned clinical language model. Performance is scoped to the evaluated cohort.



Next steps

Real ClinicalBERT-based note extraction, a live antibiogram feed for continuous retraining, clinical decision support (CDS) hooks at order entry, and exploration of whole-genome-sequencing resistance markers.

Responsible-AI design choices, not afterthoughts



Human-in-the-loop by design

Every prediction is a decision-support input reviewed by a stewardship pharmacist — the dashboard states this explicitly and permanently, not in fine print.



Honest baseline reporting

Every headline number is reported alongside the 0.50 blind baseline, never presented in isolation — no cherry-picked comparison.



Disclosed data limitations

The synthetic-cohort caveat and the regex-vs-ClinicalBERT gap are both first-class limitations in the model card, not buried assumptions.



Temporal validation before any drift claim

No stability claim is made without the rolling-origin backtest evidence to support it (Section 06).

KEY TAKEAWAYS

A rigorous, honest case for ML-assisted antimicrobial stewardship



+43.4% lift over baseline



78.2% MDRO recall



1.29% drift over 30 months



Pharmacist-gated by design

Skills demonstrated

Multi-label clinical ML • temporal / rolling-origin validation • feature engineering under leakage constraints • model documentation & responsible-AI framing • stakeholder-ready dashboarding

Anthony Chinedu Echem

Machine Learning Engineer & Data Scientist



Portfolio contact available on request

Thank you